Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

Mean and Quantiles; Left Truncation; Left Censoring; Right Truncation; Turnbull

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1 Estimating the Mean Survival Time

1.1 Mean as the area under the curve

Since $\mu = \int_0^\infty S(t)\,\mathrm{d}t$, estimate $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the last observation is censored, $\hat{S}$ never reaches 0 and the integral diverges; the estimate then depends on how the tail is completed.

1.2 First solution: Efron's completion

$\hat{S}(t) = 0$ for $t > t_{\max}$ gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$; it assumes survivors at $t_{\max}$ die immediately (biased downward), and $t_{\max}$ is random.

1.3 Second solution: restricted mean

Theorem 7.1: Restricted mean

For a pre-specified τ, estimate $\mu_\tau = \mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$ (sum over $t_j \le \tau$).

Remark 7.2: Why the weights $\int_{t_j}^{\tau} \hat{S}$

Perturbing $\hat{h}_j$ rescales $\hat{S}(t)$ for $t \geq t_j$, changing only the area right of t_j; by the delta method its variance contribution is (that area)^2 times Greenwood's term (Exercise 7.2).

$\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1} - s_k)$ over consecutive jump points s_k in $[0, \tau]$.

Example 7.3: 6-MP group (data of Lecture 5) [Supplement]

$\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6, 7, 10, 13, 16, 22, 23$ ($Y_j = 21, 17, 15, 12, 11, 7, 6$); $\tau = 35$:

$$\begin{aligned} \hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690) \\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks}, \\ \hat{V}(\hat{\mu}_{35}) &\approx 7.99, \quad \text{SE} \approx 2.83, \quad 95\% \text{ CI} \approx (17.75,\ 28.83). \end{aligned}$$

2 Quantiles and Their Confidence Intervals

$\hat{x}_p = \inf\{t : \hat{S}(t) \leq 1 - p\}$; its variance involves the hard-to-estimate $f(x_p)$, so CIs **invert** pointwise CIs for $S(t)$.

Theorem 7.4: CIs for x_p

A $100(1-\alpha)\%$ CI for x_p is the set of all t satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \leq \frac{\hat{S}(t) - (1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \leq Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left| \frac{[\log\{-\log \hat{S}(t)\} - \log\{-\log(1-p)\}]\, \hat{S}(t) \log \hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \leq Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left| \frac{2\{\arcsin\sqrt{\hat{S}(t)} - \arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \leq Z_{1-\alpha/2}.$$

Derivation: where these come from

Each is a Wald test of $H_0 : S(t) = 1 - p$ on the transformed scale, $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\text{SE}(\hat{S})) \leq Z$, with $1/|g'(\hat{S})| = |\hat{S}\log\hat{S}|$ (log-log) or $2\sqrt{\hat{S}(1-\hat{S})}$ (arcsine); collect the non-rejected t.

Example 7.5: Median for the 6-MP group

$\hat{S}(22) = 0.538 > 0.5 \geq \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. All three criteria hold from $t = 13$ on ($\hat{S} = 0.690$, SE 0.107), not for $t \leq 10$, and never fail again: 95% CI $[13, \infty)$.

3 Right-Censored and Left-Truncated Data

Definition 7.6: Data and modified risk set

Data $\{(L_i, T_i, \delta_i)\}$: entry age L_i, death or censoring time T_i. At death times t_j, d_j deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

those *entered* and still observed at t_j.

With this Y_j, KM, Nelson–Aalen, Greenwood, CIs and bands (K&M 4.2–4.4) apply unchanged.

Remark 7.7: What is being estimated

Subjects are seen only after entry, so the targets are conditional:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard is unaffected by conditioning, so d_j/Y_j still estimates it (Exercise 7.3).

Example 7.8: Channing House

Channing House residents (97 males, 365 females); truncation times are entry ages (months). The risk set is small at young ages, peaks around 950–1000 months, then falls.

- Males: the first two entered at 751, 759 and died at 777, 781; the third entered at 782.
- Plain KM: factors 1/2 at 777 ($Y = 2$) and 0 at 781 ($Y = 1$), so $\hat{S}_L(t) = 0$ for $t \ge 781$, destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \ge a)$ for an age a with a reasonable risk set:

$$\hat{S}_a(t) = \prod_{a \le t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \le t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \le t_j \le t} \frac{d_j}{Y_j}, \qquad t \ge a.$$

4 Left-Censored Data: Reversing the Time Scale

X failure time, C left censoring time; observe $T = \max(X, C)$, $\delta = I(X \ge C)$.

Theorem 7.9: Reversed time-scale trick

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \le \tau - C_i).$$

With $G(u) = P(\tau - X > u)$, KM on $\{(\tau - T_i, \delta_i)\}$ gives $\hat{G}$, and

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

Likewise for CIs and bands.

Why [Supplement]. $\max(X,C) = \tau - \min(\tau - X, \tau - C)$, $X \geq C \iff \tau - X \leq \tau - C$, and $G(\tau - u) = P(X < u)$. ■

5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ given $X_i < R_i$. With $\tau > \max_i R_i$, $\tau - X_i$ is *left truncated* by $\tau - R_i$; apply the left-truncation method to $\tau - X_i$ given $\tau - X_i > a$.

Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$

$$G^*(t) = P(\tau - X_i \geq t \mid \tau - X_i \geq \tau - \tau^*) = P(\tau - X_i \geq t \mid X_i \leq \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \leq \tau^*) = 1 - G^*(\tau - t).$$

Theorem 7.11: KM on the reversed scale

With $\tau - t_j$ the D distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \leq \tau - t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \geq \tau - t_j \geq \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$, with inference from the LT properties of $\hat{G}^*$.

Nothing in the data speaks about $X > \tau^*$.

Remark 7.12: Quick review

KM and NA extend to LT + RC data (modified risk set) and to left-censored or right-truncated data (reversed time).

6 Other Sampling Schemes

Interval and double censoring (with or without LT) need Turnbull's iterative methods (Lecture 8).

7 Selected exercises

Motivation. The identity behind μ_τ, its variance by the delta method, and why the modified risk set estimates the hazard.

Exercise 7.1 (The restricted mean)

Let $X \geq 0$ have survival function S and let $\tau > 0$.
(a) Show that $\mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$.
(b) Without censoring and with $\tau \geq X_{(n)}$, show that $\hat{\mu}_\tau = \bar{X}$.

Solution [Supplement].

(a) $\min(x,\tau) = \int_0^\tau I(t<x)\,\mathrm{d}t$; by Tonelli

$$\mathrm{E}[\min(X,\tau)] = \int_0^\tau P(X>t)\,\mathrm{d}t = \int_0^\tau S(t)\,\mathrm{d}t.$$

(b) By Exercise 5.1(a), $\hat S(t) = n^{-1}\sum_i I(X_i>t)$, zero for $t \ge X_{(n)}$:

$$\hat\mu_\tau = \frac{1}{n}\sum_{i=1}^n \int_0^\tau I(t<X_i)\,\mathrm{d}t = \frac{1}{n}\sum_{i=1}^n \min(X_i,\tau) = \bar X.$$

■

Exercise 7.2 (Variance of the restricted mean)

Derive the variance estimator $\hat V(\hat\mu_\tau)$ of this lecture by the delta method, under the same approximations as Greenwood's formula (Lecture 5): the $\hat h_j = d_j/Y_j$ are approximately uncorrelated with $\mathrm{Var}(\hat h_j) \approx \hat h_j(1-\hat h_j)/Y_j$; assume $Y_j > d_j$ for $t_j \le \tau$.

Solution [Supplement].

Step 1. $\hat S(t) = \prod_{t_j\le t}(1-\hat h_j)$, so for $t_j \le \tau$

$$\frac{\partial \hat S(t)}{\partial \hat h_j} = -\frac{\hat S(t)}{1-\hat h_j}\, I(t \ge t_j).$$

Step 2. Hence

$$\frac{\partial \hat\mu_\tau}{\partial \hat h_j} = -\frac{1}{1-\hat h_j}\int_{t_j}^\tau \hat S(t)\,\mathrm{d}t =: -\frac{A_j}{1-\hat h_j}.$$

Step 3. With uncorrelated $\hat h_j$,

$$\mathrm{Var}(\hat\mu_\tau) \approx \sum_{t_j\le\tau} \frac{A_j^2}{(1-\hat h_j)^2}\cdot\frac{\hat h_j(1-\hat h_j)}{Y_j} = \sum_{t_j\le\tau} A_j^2 \frac{\hat h_j}{Y_j(1-\hat h_j)} = \sum_{t_j\le\tau} A_j^2 \frac{d_j}{Y_j(Y_j-d_j)},$$

which is $\hat V(\hat\mu_\tau)$; with $\hat S(t)$ in place of A_j this is Greenwood's formula. ■

Exercise 7.3 (Why the modified risk set works under left truncation)

Let X be discrete with support $\{x_j\}$ and hazard $h(x_j) = P(X=x_j)/P(X\ge x_j)$. Let the entry time L and censoring time C be such that X, L, C are mutually independent; a subject is sampled only if $L \le X$ and then yields (L, T, δ) with $T = \min(X,C)$, $\delta = I(X\le C)$. Show that, for every x_j with $P(L \le x_j \le T) > 0$,

$$P(T = x_j,\ \delta = 1 \mid L \le x_j \le T,\ \text{sampled}) = h(x_j).$$

Solution [Supplement].

Step 1. $\{L \le x_j \le T\} \subseteq \{L \le X\}$ (sampling), so

$$P(\cdot \mid L \le x_j \le T,\ \text{sampled}) = P(\cdot \mid L \le x_j \le T).$$

Step 2. By independence,

$$\frac{P(X=x_j,\ C\ge x_j,\ L\le x_j)}{P(X\ge x_j,\ C\ge x_j,\ L\le x_j)} = \frac{P(X=x_j)P(C\ge x_j)P(L\le x_j)}{P(X\ge x_j)P(C\ge x_j)P(L\le x_j)} = h(x_j).$$

The numerator is $P(T=x_j, \delta=1,\ L\le x_j\le T)$, the denominator $P(L\le x_j\le T)$.

So in the modified risk set the chance of an observed death at t_j is $h(t_j)$; multiplying $1-h$ over $x_j \ge a$ gives only $P(X > t \mid X \ge a)$. ■